

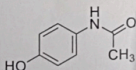
*Note: Use of calculator is allowed (not borrowing)
Attempt any two questions from each section
Attempt all parts of one question together*

	Section A	Marks
Q1	(a) What is meant by Disinfection of water? Explain how disinfection of water is carried out by using bleaching powder & chlorine.	4
	(b) Describe ion-exchange process for demineralization of water	4
	(c) Explain the basic principle of EDTA method.	2
	(d) Calculate the quantity of lime & soda required for softening of 7500 liters of water containing: $\text{CaCO}_3 = 75 \text{ mg/L}$; $\text{CaSO}_4 = 120 \text{ mg/L}$; $\text{MgCO}_3 = 20 \text{ mg/L}$; $\text{MgCl}_2 = 28 \text{ mg/L}$ and $\text{NaCl} = 3.5 \text{ mg/L}$	2.5
Q2	(a) Differentiate between lubricants & lubrication with one example of each.	2
	(b) What is the need of a lubricant, Explain.	2
	(c) Name the lubricants used in Fluid-film, Thin-film and Extreme Pressure lubrication.	3
	(d) Explain Thick-film lubrication with the help of a diagram.	3
	(e) Draw a neat & properly labelled diagram of cloud & pour point apparatus.	2.5
Q3	(a) Give reasons why corrosion takes place?	2.5
	(b) Explain the mechanism of electrochemical corrosion in detail along with the diagram.	6
	(c) Discuss any four preventive measures to control corrosion.	4

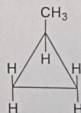
Section B

- Q4** (a) A fuel is found to contain C = 90%; H = 6.0%; S = 2.5%; O = 1.0% and ash = 0.5%. Calculate the amount of air required for complete combustion of 1 kg of fuel. If 25% excess air is used for combustion, calculate the percentage composition of the dry products of combustion. 4
- (b) How many NMR signals with multiplicity you would expect for the following structures. 4

(i)



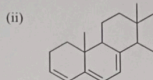
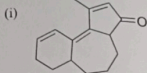
(ii)



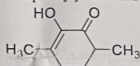
- (c) Discuss moving-bed catalytic cracking with a neat flow diagram. 3
- (d) Explain an auxochrome with suitable example. 1.5

- Q5 (a) A gaseous fuel has the following composition by volume: Methane = 5%, Hydrogen = 20%, carbon monoxide = 25%, carbon dioxide = 6% and rest nitrogen. If 20% excess air is used for combustion, then calculate volume of air supplied per m^3 of fuel and composition of dry flue gases. 4

- (b) Following the Woodward-Fieser rules, calculate the absorption maximum for each of the following compounds: 4



- (c) A fuel has high hydrogen content. Explain why its GCV is always higher than NCV, using the idea of latent heat of steam. 2
- (d) Explain anisotropic effect in shielding and deshielding of protons with example. 2.5
- Q6 (a) A sample of coal contains: C = 93%, H = 6% and ash = 1%. The following data were obtained when the above coal was tested in bomb calorimeter: Wt. of coal burnt = 0.92g; wt. of water taken = 550 g; water equivalent of bomb calorimeter = 2,200g; Rise in temperature = 2.42°C ; Fuse wire correction = 10.0 cal; Acid correction = 50.0 cal. Calculate the gross and net calorific value of coal assuming the latent heat of condensation of steam as 580 cal/g. 4
- (b) Predict λ_{max} for a $\pi \rightarrow \pi^*$ absorption band in the UV spectrum of below compound. 4



- (c) What is chemical shift? Explain how shielding and deshielding influence resonance frequency. 3.5
- (d) Which lamp is used in UV region. 1

Chromophore	Increment in nm w.r.t. the carbonyl group			
	α	β	γ	δ or higher
-OH	35	30	-	50
-OAc	6	6	6	6
-Cl	15	12	-	-
-Br	25	35	-	-
-OR	35	30	17	31
-SR	-	85	-	-
-NR ₂	-	95	-	-

MALAVIYA NATIONAL INSTITUTE OF TECHNOLOGY JAIPUR

Department of Chemistry

End-term Examination 2024-25 (Even Semester)

Engineering Chemistry-22CYT101

Duration: 150 Minutes

Date: 30-04-2025

Total Marks: 5

Note: Attempt any four questions only, and two question from each section

[Section A]

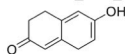
- Q.1 a. Describe zeolite process for softening of hard water. 4
b. Discuss EDTA method for the estimation of hardness present in water. 4
c. Write down the basic requirements of drinking water 2
d. Calculate the quantity of Lime and Soda required for the softening of 10000 litres of water containing: $\text{MgCO}_3=42 \text{ ppm}$; $\text{Ca}^{2+}=80 \text{ ppm}$; $\text{MgSO}_4=24 \text{ ppm}$ and $\text{CO}_2=88 \text{ ppm}$ 2.5
- Q.2 a. Define lubricants with example. Write down any six functions of lubricants. 2+3
b. Explain Extreme Pressure Lubrication mechanism. 4
c. Make a neat and labelled diagram of Flash and Fire Point apparatus. 3.5
- Q.3 a. What is corrosion? Explain by giving two examples. 3
b. Define chemical/dry corrosion. Discuss in brief the types of dry corrosion. 3.5
c. Define: 6
(i) Viscosity, (ii) Viscosity Index, (iii) Cloud and Pour Point,
(iv) Flash and Fire Point, (v) Coagulants, (vi) Disinfectants

[Section B]

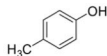
- Q.4 a. What is TMS? Draw its structure and explain. Why is it used as a reference substance in ^1H NMR? 4
b. How flue gas is analyzed using an Orsat apparatus? Explain with a neat labelled diagram. Discuss the significance of flue gas analysis. 5
c. Explain the classification of nanomaterials based on their dimensions and describe the various approaches used for their synthesis. 3.5
- Q.5 a. How nanotechnology is used to address current problems of the world? 3.5
b. A gaseous fuel has the following composition by volume: $\text{CH}_4 = 20\%$; $\text{CO} = 10\%$; $\text{CO}_2 = 5\%$; $\text{O}_2 = 2\%$; $\text{C}_2\text{H}_4 = 5\%$; $\text{C}_3\text{H}_8 = 8\%$, rest N_2 . Calculate the volume of air supplied per m^3 of fuel and the % composition of dry flue gases. 5
c. 4

Predict λ_{max} for a $\pi \rightarrow \pi^*$ absorption band in the UV spectrum of below

compound. How many NMR signals with multiplicity you would expect for the same compound. (Hint: increment for OH, $\alpha=35$, $\beta=30$, γ , δ or higher=50)



- Q.6 a. Define the terms 3
- i. Nanomaterials
 - ii. Nano Sciences
 - iii. Nanotechnology
- b. Explain the selection rules for vibrational spectroscopy? 3
- c. A sample of coal was found to have the following composition by weight 4
 C = 75%; H = 5.2%; O = 12.1%; N = 3.2% and ash = 4.5% Calculate
 (i) minimum amount of O₂ and air necessary for complete combustion of 1
 kg of coal; (ii) weight of air required if 40% excess air is supplied.
- d. What do you meant by the term 'Auxochrome'? Label the Auxochrome in 2.5
 below structure.



MALAVIYA NATIONAL INSTITUTE OF TECHNOLOGY JAIPUR
Department of Chemistry

End-term Examination 2024-25 (Odd Semester)
Engineering Chemistry-22CYT101

Date: 02-12-2024

Total Marks: 50

Duration: 150 Minutes

Note: Attempt any four questions only, and two question from each section

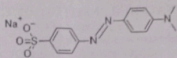
[Section A]

- Q.1 ☒ a. What is meant by disinfection of water? Explain how disinfection is carried out by using bleaching powder and chlorine. 5
- ☒ b. What are ion exchange resin? How you will purify water by using this resins? 5.5
- ☒ c. What are the advantages of this method and how ion exchangers are regenerated? 2
- ☒ d. Explain the basic principle of EDTA method? 1
- Q.2 ☒ a. Why do we need lubricants? 2
- ☒ b. Differentiate between lubricants and lubrication by giving suitable example of each. 2.5
- ☒ c. Explain Thick film Lubrication Mechanism. 2+2
- ☒ d. Explain the working of Redwood Viscometer No. 1 to measure the viscosity of lubricating oils with the help of diagram. 3
- ☒ e. What types of lubricants are being used in Thick, Thin and Extreme Pressure lubrication? 3.5
- Q.3 ☒ a. What is the reason for corrosion to take place, give two example of corrosion of metals? 4
- ☒ b. Give the mechanism of Electro-chemical Corrosion. 2
- ☒ c. Name any four methods of corrosion control. 3
- ☒ d. Calculate the amounts of Lime and Soda needed for softening 5000 litres of water containing:
 $H^+ = 4.0 \text{ mg/L}$; $MgCl_2 = 38 \text{ mg/L}$; $Al_2(SO_4)_3 = 25 \text{ mg/L}$; $Ca^{+2} = 10 \text{ mg/L}$; $HCO_3^- = 122 \text{ mg/L}$.

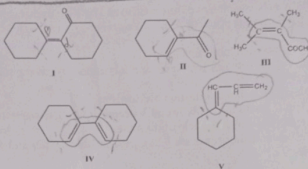
[Section B]

- Q.4 ☒ a. A fuel is found to contain C = 90%; H = 6.0%; S = 2.5%; O = 1.0% and ash = 0.5%. Calculate the amount of air required for complete combustion of 1 kg of fuel. If 25% excess air is used for combustion, calculate the percentage composition of the dry products of combustion. 4
- Or**
- A gaseous fuel has the following composition by volume – $CH_4 = 4.0\%$; $CO = 26\%$; $CO_2 = 10\%$; $H_2 = 10\%$; $N_2 = 50\%$. If 20% excess air is supplied calculate the volume of air supplied and the % composition of dry flue gases.
- ☒ b. What is knocking? How is it related to the chemical structure of the fuel? How can you reduce knocking in an internal combustion engine? 3.5
- ☒ c. What is cracking? Why it is important? Discuss the fixed bed catalytic cracking method to obtain gasoline from heavy oils. What are the limitations of the process? 5
- Q.5 ☒ a. What do you meant by the term 'Chromophore'? Label the chromophore in below structure. 2

PTO



- Five chemicals are placed in five bottles, and their labels are removed. Upon measuring the ultraviolet (UV) spectra of the contents in each bottle, values for their respective absorption maxima (λ_{max}) are obtained. Using the Woodward-Fieser rules, calculate the absorption maxima for each compound and assign the correct structures based on the observed λ_{max} values.

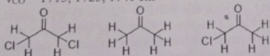


- c. Predict λ_{max} using Woodward-Fieser rules in the UV spectrum of below compound. How many NMR signals with multiplicity you would expect for the same compound.



- d. Assign the carbonyl stretching frequency of the following molecules. Justify your answer.

$$\nu_{\text{CO}} = 1715, 1725, 1740 \text{ cm}^{-1}$$



- Q.6 a. Classify nanomaterials based on their dimensions. Provide examples for each type.
- b. Explain the approaches used to synthesize nanomaterials.
- c. Explain the impact of nanotechnology in today's world.
- d. What is an Organic Photovoltaic Cell (OPV)? Explain the basic working principle of OPVs.

----- Good Luck -----

DURATION: 2.5 HOURS

MAX. MARKS 50

Attempt any four questions only and one question from the each section is compulsory

Section A

1. (a) A water sample on analysis gave the following data: $\text{Mg}(\text{HCO}_3)_2=148 \text{ ppm}$; $\text{Ca}(\text{HCO}_3)_2=124 \text{ ppm}$; $\text{MgSO}_4=90 \text{ ppm}$; $\text{CaCl}_2=111 \text{ ppm}$; $\text{HCO}_3=61 \text{ ppm}$ and $\text{Ca}(\text{NO}_3)_2=120 \text{ ppm}$. Calculate the quantity of lime (80% pure) and soda (90% pure) needed for softening 6000 litre of water. (4 marks)
- (b) Define the term scale and sludge formation in boiler. Explain the formation of scale in boiler. (3.5 marks)
- (c) What is meant by disinfection of water. Discuss any three methods for disinfection of water. (5 marks)
- 2 (a) Explain the Mechanism of thick film lubrication. (4 marks)
- (b) Draw a neat and labelled diagram of Redwood Viscometer no.1. (3 marks)
- (c) Discuss Zeolite method for softening of water. (5.5 marks)

Section B

3. (a) (i) Which of the two among Fullerene and Graphite is more reactive? Justify your statement. (ii) Compare Fullerene and Graphite in terms of electrical conductivity. (4 marks)
- (b) Draw a diagram showing the working of proton-exchange membrane fuel cells. Write down the anodic and cathode reactions involved. Write down the name of three different types of fuel cell depending on the nature of electrolyte used. (4 marks)
- (c) Calculate the oxygen balance of trinitrobenzene ($\text{C}_6\text{H}_3(\text{NO}_2)_3$) and Ammonium nitrate (NH_4NO_3). Explain why substances like trinitrotoluene (or) trinitrobenzene are mixed with ammonium nitrate in practical applications of explosives. (4.5 marks)
4. (a) Match the following process of column A with relative terms in column B (4 marks)

A	B
Fixed bed cracking	Sn and Ni catalyst
Fischer-Tropsch Process	Time Consuming
Moving bed cracking	Synthetic Fuels
Berzlius Process	Easy and more practical

- (b) A sample of coal contains: C = 80%; H = 5%; O = 1%; N = 2 %; remaining being ash. Calculate the amount of minimum air required for the complete combustion of 1 kg of coal sample. (4 marks)
- (c) Write any 3-petroleum additive that can be used instead of tetraethyl lead. Clearly mention any two advantages of an additive of your choice over the tetraethyl lead. (4.5 marks)

Section C

- (a) Relatively smaller area of anode (as compared to cathode) accelerates corrosion. Comment. (2 marks)
 - (b) Explain why a coating of Zn on Fe is advantageous over a coating of Sn on Fe. (2 marks)
 - (c) Write three important properties of slag cement. Why excess alkali reduces the strength of portland cement? (3.5 marks)
 - (d) RUL temperature is more important than refractoriness temperature. Comment. (1 mark)
 - (e) Write four furnace conditions which are responsible for failure of refractory materials. (2 marks)
 - (f) Write the approximate composition and two important applications aluminosilicate glass. (2 marks)
6. (a) With suitable polarization graph, explain how cathodic, anodic and mixed polarization of electrodes can protect a metal from corrosion. Give one example each for volatile and porous films formed on metals by reaction with gases other than oxygen. (2.5 marks)
- (b) What is the chemical composition of portland cement? How gel and crystalline hydration products provide strength to the cement paste? (3 marks)
- (c) Write the effects of dense textured brick on the properties of refractory bricks. (2 marks)
- (d) How carbon bricks are manufactured? Write two important applications of carbon bricks. (2 marks)
- (e) How safety glass is manufactured so that it doesn't fly into pieces? Write the name of glass which is used (i) as nucleating agent during glass manufacture and (ii) in fire resistant doors. (3 marks)

[illegible]

Time: 2 hours

Max.Marks: 50

Attempt any four questions. All questions carry equal marks.

Q. 1

- (a) What are disinfectants & disinfection. Explain break point chlorination and its advantages with suitable diagram. (4)
 (b) How exhausted Zeolites & ion exchange resin are regenerated (4)
 (c) Calculate the quantity of lime soda required for softening 600000 liters water containing following:
 $\text{CO}_2 = 20.00 \text{ mg/L}$, $\text{Ca}(\text{HCO}_3)_2 = 20.00 \text{ mg/L}$, $\text{Mg}(\text{HCO}_3)_2 = 25.00 \text{ mg/L}$, $\text{HCl} = 8.4 \text{ mg/L}$,
 $\text{Al}_2(\text{SO}_4)_3 = 40.00 \text{ mg/L}$, $\text{MgCl}_2 = 12.00 \text{ mg/L}$, $\text{NaCl} = 50 \text{ mg/L}$ (4.5)

Q. 2

- (a) What do understand by the term refining of crude petroleum. How is the petroleum refined. Explain with suitable diagram. Indicate the names of fractions obtained. (4.5)

(b)

What is knocking? How is it related to a chemical constitution of fuel and compression ratio of internal combustion engine. (4)

(c)

The following data were obtained when the coal sample was tested in bomb calorimeter:

Weight of coal sample = 0.95g, water taken in calorimeter = 560g, water equivalent of calorimeter = 2300g, Rise in temperature = 24.2°C , Latent heat of steam = 560 cal/g (4)

After latent heat of steam 17.71, first we have to calculate LCV & HCV of coal.

Q. 3

- (a) Write the functions of lubricants. Explain thick film lubrication and thin film lubrication with suitable diagrams. (4.5)

- (b) What are the requisites of explosives. Discuss its classification. *→ diagram* (4)

- (c) Justify, "the properties of Portland cement depends upon relative proportion of tricalcium silicates, dicalcium silicate, tricalcium aluminate." (4)

Q. 4

- (a) What is corrosion. Discuss the electrochemical corrosion (4)

- (b) How corrosion is minimized by sacrificial anodic protection method, mention its two applications with neat labeled diagrams (4.5)

- (c) What are conducting polymers, How these are prepared by doping. (4)

Q. 5

- (a) What are refractory materials. Write essential requirements of a good refractory. (4)

- (b) The selection of refractory for a furnace lining has to be chosen on its chemical nature. Explain with suitable examples. Give important requisites of an explosive? (4.5)

- (c) What is the composition of soda glass. Write its chemical & physical properties of glass. Mention the importance of annealing. (4)

Q. 6

Write short notes on ~~any~~ the following.

- (a) Disadvantages of scale formation in boilers. (4)

- (b) Hydrogen - Oxygen fuel cell & its advantages (4.5)

- (c) Draw neat labeled diagram Redwood Viscometer No-2 (4)



Time: 2 Hours

Max. Marks: 50

Attempt any four questions. All questions carry equal marks.

Discuss the structure, properties and uses of fullerenes.

Discuss the classification of explosives.

Write important applications of explosives.

Write the applications of organic electronic materials.

Discuss in detail water softening by ion-exchange method. What are advantage and disadvantage of this method?

What are the requisites of drinking water?

A water sample using $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ as a coagulant at the rate of 278 ppm, gave following data on analysis for raw water: Ca^{+2} =240 ppm; Mg^{+2} =96 ppm; CO_2 =44 ppm; HCO_3^- =732 ppm; SiO_2 =3.5 ppm; NaCl =5.85 ppm.

Calculate the lime and soda required to soften 250000 litres of water.

Explain Octane and Cetane number.

What is synthetic petrol? Describe the Fischer-Tropsch process for producing the same.

Calculate the volume of air required for complete combustion of 1 m³ of gaseous fuel having the composition: CO =44%, CH_4 =12%, H_2 =38%, C_2H_2 =4.0%, N_2 =1% and remaining being ash.

Write important characteristics of good fuel.

What is corrosion? Explain concentration cell corrosion.

Explain the following properties of lubricants: i) viscosity & viscosity index ii) flash & fire point

Discuss the chemical composition and classification of cement.

What is glass? Describe briefly the various types of glass.

Give the essential requirement of a good refractory.

Discuss Fuel cell.

Write short notes (any three)

Break point chlorination

Methods of protection against corrosion.

Caustic embrittlement

Requirements of explosives.

Classification of refractory.

MALAVIYA NATIONAL INSTITUTE OF TECHNOLOGY JAIPUR

1st B. Tech. End Term Examination (Autumn semester), 2012

Chemistry CY 101

Time: 2 Hours

Marks: 50

Attempt any five questions. All questions carry equal marks.

- ✓ 1 (a) What is caustic Embrittlement? Discuss its mechanism. 2
 (b) Explain breakpoint Chlorination by giving its relevance. 2
 (c) How will you calculate hardness by EDTA method? 3
 (d) Calculate the quantity of lime and soda required to soften 60,000 liters of hard water containing the following: $\text{Ca}(\text{HCO}_3)_2 = 20 \text{ mg/l}$; $\text{Mg}(\text{HCO}_3)_2 = 25 \text{ mg/l}$, $\text{CO}_2 = 20 \text{ mg/l}$ $\text{Al}_2\text{SO}_4 = 40 \text{ mg/l}$
 $\text{MgCl}_2 = 12 \text{ mg/l}$; $\text{HCl} = 8.4 \text{ mg/l}$. 3
- ✓ 2 (a) Discuss knocking and octane no. 3
 (b) Explain the process of reforming in detail. 3
 (c) On burning 0.83 g of a solid fuel, the temperature of 3,500 g of water increased from 26.5°C to 29.2°C . Water equivalent of calorimeter and latent heat of steam are 385.0 g and 587.0 cal/g respectively. If the fuel contains 0.7% hydrogen, calculate its gross and net Calorific value. 4
- ✓ 3 (a) Draw a neat and labeled diagram of Pensky Marten's Apparatus. 3
 (b) Discuss mechanism of lubrication. 3
 (c) Define Viscosity and Viscosity index. 4
- ✓ 4 (a) Describe the mechanism of electrochemical corrosion. Discuss the role of O_2 in concentration cell Corrosion. 6
 (b) Explain pilling bedworth rule with suitable examples. 4
- ✓ 5 (a) Discuss fullerenes on the basis of its properties and uses. 5
 (b) Give any four applications of organic electronic materials. *allow not a self-assembly body, low cost, high conductivity, so local and building* 5
- ✓ 6 (a) What are explosives? How are they classified? *chemical, nuclear, and thermite* 6
 (b) What are the uses of explosives? 4
- ✓ 7 (a) Discuss the role of Gypsum and draw a flow chart diagram of manufacturing the Portland cement. 3
 (b) Classify the type of refractories with suitable example explain pyrometric cone test of refractory. 3
 (c) Write the chemical composition of soda glass and explain the manufacture of soda glass by pot furnace method. 4

